A Primer of Algebraic D-modules

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Throughout this book, K denotes a field of characteristic zero and $K[X]$ the ring of polynomials $K[x_1, \ldots, x_n]$ in n commuting indeterminates over K.

Chapter I

Weyl Algebra

§1 Definition

Definition 1.1. *The ring $K[X]$ is a vector space of infinite dimension over K. Its algebra of K-linear operators is denoted by $\mathrm{End}_K(K[X])$. Let $\hat{x}_1, \ldots, \hat{x}_n$ be the operators of $K[X]$ which are defined on a polynomial $f \in K[X]$ by the formulae $\hat{x}_i(f) = x_i \cdot f$. Similarly, $\partial_1, \ldots, \partial_n$ are the operators defined by $\partial_i(f) = \partial f / \partial x_i$. These are linear operators of $K[X]$.*

*The n-**th Weyl algebra** A_n is the K-subalgebra of $\mathrm{End}_K(K[X])$ generated by the operators $\hat{x}_1, \ldots, \hat{x}_n$ and $\partial_1, \ldots, \partial_n$. For the sake of consistency, we write $A_0 = K$.*

Proposition 1.2.

$$[\partial_i, \hat{x}_j] = \delta_{ij} \cdot 1$$
$$[\partial_i, \partial_j] = [\hat{x}_i, \hat{x}_j] = 0$$
$$[\partial_i, f] = \frac{\partial f}{\partial x_i}$$

Proposition 1.3. *We have*

1. $$\partial^\beta x^\alpha = \sum_{\substack{\beta_1+\beta_2=\beta \\ \beta_i \in \mathbb{N}^n}} \binom{\beta}{\beta_1} \frac{\partial x^\alpha}{\partial x^{\beta_1}} \cdot \partial^{\beta_2}$$

   thus $\partial^\beta x^\alpha = x^\alpha \partial^\beta + D'$ where $\deg D' \leq |\alpha| + |\beta| - 2$

2. *The set $\mathbf{B} = \left\{x^\alpha \partial^\beta : \alpha, \beta \in \mathbb{N}^n\right\}$ is a basis of A_n as a vector space over K.*

Theorem 1.4. *Let the free K-algebra $K\{z_1, \ldots, z_{2n}\}$ and J be the two-sided ideal generated by $[z_{i+n}, z_i] - 1$ for $i = 1, 2, \ldots, n$ and $[z_i, z_j]$ for $j \neq i + n$ and $1 \leq i, j \leq 2n$. We may define a surjective homomorphism $\phi : K\{z_1, \ldots, z_{2n}\} \to A_n$ by $\phi(z_i) = x_i$ and $\phi(z_{i+n}) = \partial_i$, for $i = 1, 2, \ldots, n$.*

$$\begin{tikzcd}
\{z_1, \ldots, z_n\} \arrow[r] \arrow[d, "\iota"] & A_n \\
K\{z_1, \ldots, z_n\} \arrow[ur, "\phi"] &
\end{tikzcd}$$

It follows that $J \subseteq \ker \phi$. *Thus* ϕ *induces a homomorphism* $\bar{\phi} : K\{z_1, \ldots, z_{2n}\}/J \longrightarrow A_n$.

Proof. We may use the relations to show that every element of $K\{z_1, \ldots, z_{2n}\}/J$ may be written as a linear combination of monomials of the form

$$z_1^{m_1} \ldots z_{2n}^{m_{2n}} + J$$

by 1.3 the images of these monomials under $\bar{\phi}$ form a basis of A_n as a vector space over K. In particular, the monomials must be linearly independent in $K\{z_1, \ldots, z_{2n}\}/J$. Hence $\bar{\phi}$ is an isomorphism of vector spaces and, a fortiori, an isomorphism of rings. □

Corollary 1.5. *Let* $m < n$ *be positive integers. Choose polynomials* $f_i \in K[X]$, *for* $1 \leq i \leq n$, *as follows: if* $i \leq m$, *then* f_i *is a polynomial in the variables* $x_{m+1}, \ldots, x_n$; *otherwise* $f_i = 0$. *The map* $\sigma : A_n \longrightarrow A_n$ *defined by the formulae*

$$\sigma(x_i) = x_i + f_i$$
$$\sigma(\partial_i) = \partial_i - \sum_1^n \frac{\partial f_k}{\partial x_i} \partial_k$$

induce an automorphism of A_n.

Proof. Define a homomorphism ϕ of $K\{z_1, \ldots, z_{2n}\}$ to A_n by $\phi(z_i) = x_i + f_i$ and

$$\phi(z_{i+n}) = \partial_i - \sum_1^n \frac{\partial f_k}{\partial x_i} \partial_k$$

Choose i, j such that $1 \leq i, j \leq n$. It is clear that $\phi([z_i, z_j]) = 0$.

Let us calculate $\phi([z_{i+n}, z_j])$. Since ϕ is a ring homomorphism, this is the same as $[\phi(z_{i+n}), \phi(z_j)]$, which equals

$$\begin{aligned}
\phi([z_{i+n}, z_j]) &= [\phi(z_{i+n}), \phi(z_j)] \\
&= \left[\partial_i - \sum_1^n \frac{\partial f_k}{\partial x_i} \partial_k, x_j + f_j\right] \\
&= \delta_{ij} + \frac{\partial f_j}{\partial x_i} - \frac{\partial f_j}{\partial x_i} - \sum_1^n \frac{\partial f_k}{\partial x_i} \cdot \frac{\partial f_j}{\partial x_k} \\
&= \delta_{ij}
\end{aligned}$$

A similar calculation shows that $\phi([z_{i+n}, z_{j+n}]) = 0$. Thus ϕ induces an endomorphism $\sigma = \bar{\phi}$ of A_n.

A similar argument shows that the map τ defined by the formulae

$$\tau(x_i) = x_i - f_i$$
$$\tau(\partial_i) = \partial_i + \sum_1^n \frac{\partial f_k}{\partial x_i} \partial_k$$

is an endomorphism of A_n. It is now easy to check that τ is the inverse of σ. Thus σ is an automorphism of A_n. $\square$

§2 Degree

Definition 2.1. *Let $D \in A_n$. The **degree** of D is the largest length of the multi-indices $(\alpha, \beta) \in \mathbb{N}^n \times \mathbb{N}^n$ for which $x^\alpha \partial^\beta$ appears with non-zero coefficient in the canonical form of D. It is denoted by* $\deg(D)$. *As with the degree of a polynomial, we use the convention that the zero polynomial has degree $-\infty$.*

Lemma 2.2. *We have*

$$\partial^\beta x^\alpha = \sum_{\substack{\beta_1+\beta_2=\beta \\ \beta_i \in \mathbb{N}^n}} \partial^{\beta_1}(x^\alpha)\partial^{\beta_2}$$

thus $\partial^\beta x^\alpha = x^\alpha \partial^\beta + D'$ where $\deg D' \leq |\alpha| + |\beta| - 2$

Theorem 2.3. *The degree satisfies the following properties; for $D, D' \in A_n$:*

1. $\deg(D + D') \leq \max\{\deg(D), \deg(D')\}$. *if* $\deg(D) \neq \deg(D')$ *then we have equality in the above formula.*
2. $\deg(DD') = \deg(D) + \deg(D')$.
3. $\deg[D, D'] \leq \deg(D) + \deg(D') - 2$.

Corollary 2.4. *The algebra A_n is a domain.*

§3 Ideal Structure

Theorem 3.1. *The algebra A_n is simple.*

Proof. Let I be a non-zero two-sided ideal of A_n. Choose an element $D \neq 0$ of smallest degree in I. If D has degree 0 , it is a constant, and $I = A_n$. Assume that D has degree $k > 0$ and let us aim at a contradiction.

Suppose that (α, β) is a multi-index of length k. If $x^\alpha \partial^\beta$ is a summand of D with non-zero coefficient and $\beta_i \neq 0$, then $[x_i, D]$ is non-zero and has degree $k - 1$. Since I is a two-sided ideal of A_n, we have that $[x_i, D] \in I$. But this contradicts the minimality of D. Thus $\beta = (0, \ldots, 0)$. Since $k > 0$, we must have that $\alpha_i \neq 0$, for some $i = 1, 2, \ldots, n$. Hence $[\partial_i, D]$ is a non-zero element of I of degree $k - 1$, and once again we have a contradiction. $\square$

Corollary 3.2. *Every (ring, algebra) endomorphism of A_n is injective.*

Theorem 3.3. *Every left ideal of A_n is generated by two elements.*

Chapter II

Rings of Differential Operators

Let R be a commutative K-algebra.

§1 Definition

Definition 1.1. *Let R be a commutative K-algebra.*

(i) We will identify an element $a \in R$ with the operator of $\mathrm{End}_K(R)$ defined by the rule $r \mapsto ar$, for every $r \in R$. An operator $P \in \mathrm{End}_K(R)$ has order zero if $[a, P] = 0$, for every $a \in R$.

(ii) Suppose we have defined operators of order $< n$. An operator $P \in \mathrm{End}_K(R)$ has order n if it does not have order less than n and $[a, P]$ has order less than n for every $a \in R$.

*Let $D_n(R)$ denote the K-vector space of all operators of $\mathrm{End}_K(R)$ of order $< n$. The **ring of differential operators** of R is defined as the subring $D(R) = \bigcup_{n \geq 0} D_n(R)$.*

Remark. *It follows that $D_n(R) = \{P \in \mathrm{End}_K R : [\cdots[[P, a_1], a_2], \cdots, a_{n+1}] = 0$ for any $a_i \in R\}$ and the set of order n is $D_n - D_{n-1}$.*

Definition 1.2. *A **derivation** of the K-algebra R is a K-linear operator D of R which satisfies Leibniz's rule:*

$$D(ab) = aD(b) + bD(a) \quad \text{for every } a, b \in R$$

Let $\mathrm{Der}_K(R)$ denote the K-vector space of all derivations of R. If $D \in \mathrm{Der}_K(R)$ and $a \in R$, we define a new derivation aD by $(aD)(b) = aD(b)$ for every $b \in R$. The K-vector space $\mathrm{Der}_K(R)$ is a left R-module for this action.

Proposition 1.3. *Let R be a commutative K-algebra.*

1. *The elements of order zero are the elements of R i.e. $D^0(R) = R$*

2. *The operators of order ≤ 1 correspond to the elements of $\mathrm{Der}_K(R) + R$ .i.e.*

$$D^1(R) \subset \mathrm{Der}_K(R) + R$$

Proof. 2. Let $Q \in D^1(R)$ and put $P = Q - Q(1)$. Note that $P(1) = 0$ and that P has order ≤ 1. Hence $[P, a]$ has order zero for every $a \in R$. Thus for every $b \in R$, we have that $[[P, a], b] = 0$. Writing the commutators explicitly, one obtains the equality

$$(Pa)b - (aP)b - b(Pa) + b(aP) = 0$$

Applying this operator to $1 \in R$, we end up with $P(ab) = aP(b) + bP(a)$, it follows that P is a derivation of R. But $Q = P + Q(1) \in \mathrm{Der}_K(R) + R$, as required. □

Proposition 1.4. *Let $P \in D^n(R)$ and $Q \in D^m(R)$, then $P \cdot Q \in D^{n+m}(R)$.*

Proof. The proof is by induction on $m + n$. If $m + n = 0$ the result is obvious. Suppose the result true whenever $m + n < k$. If $m + n = k$ and $a \in R$, we have that

$$[PQ, a] = P[Q, a] + [P, a]Q$$

The definition of order implies that $[Q, a] \in D^{m-1}(R)$ and $[P, a] \in D^{n-1}(R)$. Thus, by the induction hypothesis $P[Q, a], [P, a]Q \in D^{n+m-1}$. Hence $[PQ, a]$ belongs to D^{n+m-1}, as required. □

§2 The Weyl Algebra

Proposition 2.1. *Every derivation of $K[X]$ is of the form $\sum_1^n f_i \partial_i$, for some $f_1, \ldots, f_n \in K[X]$.*

Lemma 2.2. *Let $P \in D(K[X])$. If $[P, x_i] = 0$ for every $i = 1, \ldots, n$, then $P \in K[X]$.*

Definition 2.3. *Define C_r to be the set of operators in A_n which can be written in the form $\sum_\alpha f_\alpha \partial^\alpha$ with $|\alpha| \leq r$. A simple calculation shows that*

$$C_r = C_{r+1} \cap D^r(K[X])$$

By Proposition 1.3, we have that $C_1 = \mathrm{Der}_K(K[X]) + K[X]$ and that $C_0 = K[X]$. We will use the convention that if $k < n$ then $\mathbb{N}^k$ is embedded in $\mathbb{N}^n$ as the set of n-tuples whose last $n - k$ components are zero.

Lemma 2.4. *It follows from the identity $\left[\partial^\beta, x_n\right] = \beta_n \partial^{\beta - e_n}$ that $\left[\partial^\beta, x_n\right] = 0 \Leftrightarrow \beta_n = 0$. Thus $[G, x_n] = 0$ implies that G can be written as a linear combination of monomials of the form $x^\alpha \partial^\beta$ with $\beta \in \mathbb{N}^{n-1}$.*

Lemma 2.5. *Let $P_1, \ldots, P_n \in C_{r-1}$ and assume that $[P_i, x_j] = [P_j, x_i]$ whenever $1 \leq i, j \leq n$. Then there exists $Q \in C_r$ such that $P_i = [Q, x_i]$, for $i = 1, \ldots, n$.*

Proof. Suppose, by induction, that we have determined $Q' \in C_r$ such that $[Q', x_i] = P_i$ for $k+1 \leq i \leq n$. Write $G = [Q', x_k] - P_k$, then

$$G = \sum_{\alpha \in \mathbb{N}^k} f_\alpha \partial^\alpha.$$

since $[G, x_i] = [[Q', x_k], x_i] - [P_k, x_i] = [[Q', x_i], x_k] - [P_k, x_i] = 0$ for $k+1 \leq i \leq n$.

Now write

$$Q'' = \sum_{\alpha \in \mathbb{N}^k} (\alpha_k + 1)^{-1} f_\alpha \partial^{\alpha + e_k} \in C^r$$

We have $[Q'', x_k] = G$ and $[Q'', x_i] = 0$. Thus $[Q' - Q'', x_i] = P_i$, for $k \leq i \leq n$; and the induction is complete. □

Theorem 2.6. *The ring of differential operators of $K[X]$ is $A_n(K)$ i.e. . Besides this, .*

1. $D^k(K[X]) = C_k$
2. $D(K[X]) = A_n(K)$

Proof. It is enough to prove that $D^k(K[X]) \subseteq C_k$. Let $P \in D(K[X])$. If $P \in D^1(K[X])$ then by Lemma 1.1, $P \in \mathrm{Der}_K(K[X]) + K[X]$. Thus $P \in C_1$ by Proposition 1.3.

Suppose, by induction, that $D^k(K[X]) = C_k$ for $k \leq m-1$. Let $P \in D^m(K[X])$. Write $P_i = [P, x_i]$. Since P_i has order $k \leq m-1$, it follows that $P_i \in C_{n-1}$. But, for all $1 \leq i, j \leq n$,

$$[P_i, x_j] = [[P, x_i], x_j] = [[P, x_j], x_i] = [P_j, x_i].$$

Thus by Lemma 2.2 there exists $Q \in C_m$ such that $[Q, x_i] = P_i, 1 \leq i \leq n$. Hence $[Q - P, x_i] = 0$ in $D(K[X])$. Since this holds whenever $1 \leq i \leq n$, we conclude by Lemma 2.1 that $Q - P \in K[X] = C_0$. Therefore $P \in C_m$. Hence $D^m(K[X]) \subseteq C^m$, as we wanted to prove. □

Theorem 2.7. $\mathrm{ord}\,(PQ) = \mathrm{ord}(P) + \mathrm{ord}(Q)$

Chapter III

Jacobian Conjecture

§1 Polynomial Maps

Definition 1.1. *Let $F : K^n \to K^m$. We say that F is **polynomial** if there exist $F_1, \ldots, F_m \in K[x_1, \ldots, x_n]$ such that $F(p) = (F_1(p), \ldots, F_m(p))$ for all $p \in K^n$.*

*A polynomial map is called an isomorphism or a **polynomial isomorphism** if it has an inverse which is also a polynomial map.*

For the rest of the section we shall write X, Y for the spaces K^n and K^m; and $K[X], K[Y]$ for the polynomial rings $K[x_1, \ldots, x_n]$ and $K[y_1, \ldots, y_m]$.

Definition 1.2. *Suppose that $F : X \to Y$ is a polynomial map. We may define a map,*

$$F^\sharp : K[Y] \to K[X], \quad \text{given that } g \mapsto g \circ F$$

*The map $F^\sharp$ is called the **comorphism** of F.*

Suppose that a ring homomorphism $\phi : K[Y] \to K[X]$ is given. Then we may use it to construct a polynomial map from X to Y. Now let

$$\phi_\sharp : X \to Y, \quad \mathbf{x} \mapsto (\phi(y_1)(\mathbf{x}), \ldots, \phi(y_m)(\mathbf{x}))$$

Theorem 1.3. *Let $F : X \to Y$ and $G : Y \to Z$ be polynomial maps, then*

1. $\left(F^\sharp\right)_\sharp = F$
2. $G \cdot F : X \to Z$ *is a polynomial map and* $(G \cdot F)^\sharp = F^\sharp \cdot G^\sharp$

Theorem 1.4. *If $\phi : K[Y] \to K[X]$ and $\psi : K[Z] \to K[Y]$ are homomorphism of polynomial rings, then*

1. $(\phi_\sharp)^\sharp = \phi$.
2. $(\phi \cdot \psi)_\sharp = \psi_\sharp \cdot \phi_\sharp$.

Corollary 1.5. *A polynomial map $F : X \to Y$ is an isomorphism if and only if $F^\sharp$ is an isomorphism.*

§2 Jacobian Conjecture

Jacobian conJecture. Let $F : K^n \to K^n$ be a polynomial map. If $\Delta F = 1$ on K^n then F has a polynomial inverse on the whole of K^n.

Lemma 2.1. *Let $F : X \to X$ be a polynomial map and suppose that $\Delta F \neq 0$ everywhere in X. Then $F^\sharp$ is injective.*

Proof. Suppose that $F^\sharp$ is not injective, and choose the non-constant polynomial $g \in K[X]$ of smallest degree such that $F^\sharp(g) = 0$. Then $g \circ F = 0$. Let $g_i = \partial g / \partial x_i$ and

$$\mathbf{v} = (g_1(F_1, \ldots, F_n), \ldots, g_n(F_1, \ldots, F_n)).$$

Hence, by the chain rule,

$$D(g \circ F) = \mathbf{v}(p) \cdot JF(p) = 0$$

for every $p \in X$. Since

$$\Delta F(p) = \det JF(p) \neq 0,$$

we conclude that $\mathbf{v}(p) = 0$ for every $p \in X$. Thus $g_i(F_1, \ldots, F_n) = 0$ for $1 \leq i \leq n$. Since g is not constant, at least one of the g_i must be non-zero. But g_1 has degree smaller than g, a contradiction. □

Proposition 2.2. *Denote by $K[F_1, \ldots, F_n]$ the subalgebra of $K[X]$ generated by the coordinate functions of F. This is the image of the homomorphism $F^\sharp$. Thus the Jacobian conjecture may be rephrased as follows.*

Let $F : K^n \to K^n$ be a polynomial map and assume that $\Delta F = 1$ in K^n. Then $K[F_1, \ldots, F_n] = K[x_1, \ldots, x_n]$.

§3 Derivation

Definition 3.1. *Let D be a derivation of a K-algebra S.*

1. *It follows from Leibniz's rule that the kernel of D is a subring (subalgebra) of S, it is called the **ring of constants of** D.*

2. *The derivation D is **locally nilpotent** if given $a \in S$, there exists $k \in \mathbb{N}$ such that $D^k(a) = 0$.*

3. *Let S be a ring and D a locally nilpotent derivation. Define a map $\phi : S \longrightarrow S[x]$ by the rule*

$$\phi(a) = \sum_0^\infty \frac{D^n(a)}{n!} x^n$$

for every $a \in S$. It is easy to check that ϕ is a ring homomorphism which satisfies

$$\phi \cdot D = \partial \cdot \phi$$

Proposition 3.2. *Let S be a K-algebra and $D_1, \ldots, D_n$ be commuting locally nilpotent derivations of S. Suppose that there exist $t_1, \ldots, t_n \in S$ such that $D_i(t_j) = \delta_{ij}$. Then*

1. *$S = R[t_1, \ldots, t_n]$, where R is the ring of constants with respect to $D_1, \ldots, D_n$,*
2. *$t_1, \ldots, t_n$ are algebraically independent over R,*
3. *$D_i = \partial/\partial t_i$ for $i = 1, \ldots, n$.*

Proof. Proof: We firstly prove it whene $n = 1$. Put $\bar{S} = S/St$. Let $\rho : S \longrightarrow \bar{S}[x]$ be the composition of ϕ defined above and the projection $S[x] \longrightarrow \bar{S}[x]$. We want to show that ρ is an isomorphism. Note that $\rho(t) = x$.

To prove that ρ is surjective it is enough to prove that its image contains $\bar{S}$. Let $a \in S$. Denote by $\bar{a}$ its image in $\bar{S}$. Since D is locally nilpotent, there exists $n \in \mathbb{N}$ such that $D^k(a) = 0$ for $k > n$. Thus,

$$\rho(a) = \sum_0^n \frac{\overline{D^i(a)}}{i!} x^i.$$

If $n = 0$, then $\rho(a) = \bar{a}$. If $n > 0$ put $a_0 = a$ and define $a_{j+1} = a_j - D^{n-j}(a_j)\, t^{n-j}/(n-j)!$, for $j = 1, \ldots, n$. It is easy to show, by induction on j, that $D^k(a_j) = 0$ for $k > n - j$ and that

$$\rho(a_j) = \sum_0^{n-j} \frac{\overline{D^i(a_j)}}{i!} x^i.$$

Thus $\rho(a_n) = \bar{a}_n$. However, since $\bar{t} = 0$, we have that $\rho(a_n) = \bar{a}$. Thus ρ is surjective.

Let us prove that ρ is injective. If not, then there exists a non-zero $a \in S$ such that $\rho(a) = 0$. Thus $D^k(a) \in tS$, for every $k \in \mathbb{N}$. Hence $a = a_1 \cdot t$, for some $a_1 \in S$. Since $\rho(t) = x$, we have that $\rho(a_1) = 0$. Thus $a_1 \in tS$ and $a = a_2 \cdot t^2$, for some $a_2 \in S$. Continuing this way we conclude that t^n divides a for all $n \geq 0$. But this is impossible, unless $a = 0$. Indeed, ϕ maps t to $t + x$. Thus if t^n divides a, we also have that $\phi(t^n) = (t + x)^n$ divides $\phi(a)$ in the polynomial ring $S[x]$. Hence, if $a \neq 0$ we have that $\deg(\phi(a)) \geq n$ for every $n > 0$, which is clearly impossible. Thus $a = 0$, as required.

We conclude that the homomorphism $\rho : S \longrightarrow \bar{S}[x]$ is an isomorphism. Since $\rho \cdot D = d/dx \cdot \rho$, we have that $R = \rho^{-1}(\bar{S})$. The result now follows if we recall that $\rho(t) = x$.

We proceed by induction on the number n of derivations. By Lemma 3.2, $S = R_1[t_1]$, where R_1 is the ring of constants of D_1. But t_1 is algebraically independent over R_1 and $D_1 = d/dt_1$. Since D_1 commutes with D_i for $i > 1$, we have that $D_i(R_1) \subseteq R_1$. Thus, by the induction hypothesis, $R_1 = R[t_2, \ldots t_n]$, and the proposition follows. $\square$

§4 Automorphisms

Definition 4.1. *Let $X = K^n$. The rational function field of $K[X]$ will be denoted by $K(X)$. Let $F : X \to X$ be a polynomial map with coordinate functions $F_1, \ldots, F_n$. Assume that*

$$\Delta = \Delta F \neq 0$$

everywhere on X.

1. *Define a map $D_i : K(X) \to K(X)$ by*

$$D_i(g) = \Delta^{-1} \det J\left(F_1, \ldots, F_{i-1}, g, F_{i+1}, \ldots, F_n\right).$$

   It is easy to check that D_i is a derivation of $K(X)$.

2. *Now let $K\left[X, \Delta^{-1}\right]$ be the K-subalgebra of $K(X)$ of all rational functions whose denominator is a power of Δ. Then D_i restricts to a derivation of $K\left[X, \Delta^{-1}\right]$, since $D_i\left(\Delta^{-1}\right) = -\Delta^2 D_i(\Delta) \in K\left[X, \Delta^{-1}\right]$*

Proposition 4.2. *Let R be a commutative ring and D, D' be derivations of R, then $[D, D']$ is a derivation of R.*

Proposition 4.3. *Let D be a K-derivation of $K\left[x_1, \ldots, x_n\right]$.*

1. *D can be extended to the power series ring $K\left[\left[x_1, \ldots, x_n\right]\right]$.*

2. *If Δ is a power series such that $\Delta(0) \neq 0$ then $\Delta^{-1} \cdot D$ is a derivation of the power series ring $K\left[\left[x_1, \ldots, x_n\right]\right]$.*

Lemma 4.4. *As derivations of $K\left[X, \Delta^{-1}\right]$ the D_i satisfy:*

1. *$D_i\left(F_j\right) = \delta_{ij}$.*

2. *The D_i commute pairwise.*

Proof. Note first that $\Delta(0) \neq 0$. Thus Δ is invertible as a power series and $K\left[X, \Delta^{-1}\right] \subseteq K[[X]]$. On the other hand, $\Delta \cdot D_i$ is a derivation of $K\left[x_1, \ldots, x_n\right]$ which can be extended to a derivation on the power series ring $K[[X]] = K\left[\left[x_1, \ldots, x_n\right]\right]$. Since Δ is invertible as a power series, then D_i can also be extended to a derivation of $K[[X]]$.

Put derivation $B = [D_i, D_j]$. We want to show that $B = 0$ on $K\left[X, \Delta^{-1}\right]$. It is enough to show that $B = 0$ on the power series ring $K[[X]]$.

Moreover $B\left(F_k\right) = 0$, for $1 \leq k \leq n$; and so B is zero in the subalgebra $K\left[F_1, \ldots, F_n\right]$. But $F_1, \ldots, F_n$ are algebraically independent, by Lemma 2.2. Hence we may consider B as a derivation on the power series ring $K\left[\left[F_1, \ldots, F_n\right]\right]$. By (1), B is zero on $K\left[\left[F_1, \ldots, F_n\right]\right]$.

For $1 \leq i \leq n$ let $a_i = F_i(0)$. The jacobian matrices of $(F_1 - a_1, \ldots, F_n - a_n)$ and F coincide. Since the latter is invertible in $K[[x_1, \ldots, x_n]]$, we conclude from the local inversion theorem (see Appendix 2) that

$$K[[x_1, \ldots, x_n]] = K[[F_1 - a_1, \ldots, F_n - a_n]] = K[[F_1, \ldots, F_n]].$$

Thus B is zero on $K[[x_1, \ldots, x_n]]$, as required. □

Definition 4.5. *Let $a \in A_n$. The map $\mathrm{ad}_a : A_n \to A_n$ is defined by*

$$\mathrm{ad}_a(b) = [a, b].$$

This is a K-linear map, but it is not a K-algebra homomorphism.

Theorem 4.6. *Let $F : K^n \to K^n$ be a polynomial map and assume that $\Delta F = 1$ everywhere on K^n. If every endomorphism of A_n is an automorphism, then the Jacobian conjecture holds.*

Proof. Since $\Delta F = 1$, it follows from Lemma 4.1 that $D_1, \ldots, D_n$ are derivations of $K[X]$ which satisfy

$$[D_i, F_j] = D_i(F_j) = \delta_{ij} \text{ and } [D_i, D_j] = 0$$

for $1 \leq i, j \leq n$. By, there exists an endomorphism $\phi : A_n \to A_n$ such that $\phi(x_i) = F_i$ and $\phi(\partial_i) = D_i$, for $1 \leq i \leq n$. Note that for $b \in A_n$,

$$\deg(\mathrm{ad}_{\partial_i}(b)) = \deg[\partial_i, b] \leq \deg b - 1$$

Thus given $b \in A_n$, there exists $k \in \mathbb{N}$ such that $(\mathrm{ad}_{\partial_1})^k(b) = 0$. Since

$$\phi(\mathrm{ad}_{\partial_i}(b)) = \mathrm{ad}_{D_i}\phi(b)$$

we have that $(\mathrm{ad}_{D_i})^k(\phi(b)) = 0$. Assuming that ϕ is an automorphism, we conclude that D_i is locally nilpotent. It then follows by Proposition 3.1 that $K[F_1, \ldots, F_n] = K[x_1, \ldots, x_n]$, which is the Jacobian conjecture as stated in 2.3. □

Chapter IV

Modules Over The Weyl Algebra

§1 The Polynomial Ring

Proposition 1.1. *Let R be a ring and M an irreducible left R-module.*

1. $M = Rm$ *for every* $0 \neq m \in M$.
2. *If* $0 \neq u \in M$, *then* $M \cong R/ann_R(u)$.
3. *If R is not a division ring, then M is a torsion module.*

Proposition 1.2. *The A_n-module $K[X]$ is an irreducible, torsion A_n-module. Besides this, we have isomorphism of A_n-module*

$$K[X] \cong A_n / \sum_1^n A_n \partial_i.$$

Proof. First of all 1 is clearly a generator of A_n-module $K[X]$ and the annihilator of 1 is the left ideal generated by $\partial_1, \ldots, \partial_n$.

$$\begin{tikzcd} A_n \arrow[r] \arrow[d] & K[X] \\ A_n / \sum_1^n A_n \partial_i \arrow[ur] & \end{tikzcd}$$

□

Remark. *Choose $g_1, \ldots, g_n \in K[X]$ and consider the left ideal J of A_n generated by $\partial_1 - g_1, \ldots, \partial_n - g_n$. It is easy to check that the map*

$$\psi : A_n/J \longrightarrow K[X] \text{ defined by } \psi(f + J) = f$$

is an isomorphism of A_n-modules.

Proposition 1.3. *Another A_n-module is $A_n / \sum_1^n A_n \cdot x_i$.*

$$\overline{x^\alpha \partial^\beta} = (-1)^{|\alpha|} \, \overline{\partial^{\beta - \alpha}}$$

As a K-vector space it is isomorphic to $K[\partial] = K[\partial_1, \ldots, \partial_n]$, the set of polynomials in $\partial_1, \ldots, \partial_n$. Using this isomorphism, we may identify the action of A_n directly on $K[\partial]$: the ∂ 's act by multiplication, whilst x_i acting on ∂_j gives $-\delta_{ij} \cdot 1$.

§2 Twisting

Definition 2.1. *Let R be a ring and M a left R module. Suppose that σ is an automorphism of R. We shall define a new left module M_σ, as follows. As an abelian group, $M_\sigma = M$. Let $a \in R$ and $u \in M$, define*

$$r \cdot m = \sigma(r)m$$

*It is called the **twisted module of** M **by** σ.*

Proposition 2.2. *Let R be a ring, M a left R-module and σ an automorphism of R. Then:*

1. *M_σ is irreducible if and only if M is irreducible.*
2. *M_σ is a torsion module if and only if M is a torsion module.*
3. *$M_\sigma \oplus M'_\sigma \cong (M \oplus M')_\sigma$.*
4. *If N is a submodule of M then $(M/N)_\sigma \cong M_\sigma/N_\sigma$.*
5. *Let J be a left ideal of R. Set $\sigma(J) = \{\sigma(r) : r \in J\}$. Then $\sigma(J)$ is a left ideal of J and $(R/J)_\sigma \cong R/\sigma^{-1}(J)$.*

Proposition 2.3. *The Fourier transform of $K[X]$ is $K[\partial]$.*

Proof. Let $\mathcal{F}$ be the Fourier transform automorphism of A_n. By Proposition 5.2, the twisted module $(K[X])_{\mathcal{F}}$ is isomorphic to

$$A_n / \sum_1^n A_n \cdot \mathcal{F}^{-1}(\partial_i) = A_n / \sum_1^n A_n \cdot x_i \cong K[\partial]$$

□

Theorem 2.4. *For every positive integer r let σ_r be the automorphism of A_n which satisfies $\sigma_r(x_i) = x_i$ and $\sigma_r(\partial_i) = \partial_i - x_i^r$. The modules $K[X]_{\sigma_r}$ form an infinite family of pairwise nonisomorphic irreducible modules over A_n.*

Proof. Let $r < t$, and suppose that there exists an isomorphism, $\phi : K[X]_{\sigma_r} \longrightarrow K[X]_{\sigma_t}$. Since $K[X]_{\sigma_r}$ is irreducible, it is generated by 1 . Thus ϕ is completely determined by the image of 1 ; say $\phi(1) = f \neq 0$. Now the equation $\phi(\partial_i \bullet 1) = \partial_i \bullet \phi(1)$ translates as the differential equation

$$\phi(\partial_i \bullet 1) = \phi((\partial_i - x_i^r)(1)) = \phi(-x_i^r) = -\phi(x_i^r) \text{ and } \partial_i \bullet \phi(1) = (\partial_i - x^r)(f)$$

thus

$$\frac{\partial f}{\partial x_i} = \left(x_i^t - x_i^r\right) f.$$

The left hand side of the equation has degree $\leq \deg f - 1$. Since $f \neq 0$ and $r < t$, the right hand side has degree $\deg f + t$. This is a contradiction, so the theorem is proved. □

§3 Holomorphic Functions

Lemma 3.1. *Let $h(z)$ be the holomorphic function* $\exp(\exp(z))$. *For every positive integer m there exists a polynomial $F_m(x) \in \mathbb{C}[x]$ of degree m such that*

$$d^m h/dz^m = F_m\left(e^z\right) h(z).$$

Proof. The proof is by induction on m. If $m = 1$ then $dh/dz = e^z h(z)$, so we may take $F_1(x) = x$. Suppose that the result is true for $m = k$. Then

$$d^{k+1}h/dz^{k+1} = d/dz\left(F_k\left(e^z\right) h(z)\right) = \left(e^z F_k'\left(e^z\right) + e^z F_k\left(e^z\right)\right) h(z).$$

Thus we may take

$$F_{k+1}(x) = xF_k'(x) + xF_k(x)$$

which is a polynomial of degree $k + 1$. □

Proposition 3.2. *The function $h(z) =$* $\exp(\exp(z))$ *is not a torsion element of the $A_1(\mathbb{C})$-module $\mathcal{H}(U)$.*

Proof. Suppose that there exists a non-zero operator $P \in A_1(\mathbb{C})$ such that $P \cdot h = 0$. Write

$$P = \sum_{i=0}^{m} f_i(z)\partial^i$$

with $f_m(z) \neq 0$. By Lemma 6.1, we have

$$0 = P \cdot h = \sum_{i=0}^{m} f_i(z) F_i\left(e^z\right) h(z).$$

Since $h(z) \neq 0$, we conclude that

$$\sum_{i=0}^{m} f_i(z) F_i\left(e^z\right) = 0.$$

But this is impossible since F_m has degree m and $f_m(z) \neq 0$. □

Chapter V

Differential Equations

§1 The D-Modules of Equations

Definition 1.1. *Let P be an operator in A_n. This differential operator gives rise to the equation*

$$P(f) = \sum_{\alpha} g_\alpha \partial_\alpha(f) = 0 \text{ in } K[X],$$

where $f \in K[X]$. More generally, if $P_1, \ldots, P_m$ are differential operators in A_n, then we have a system of differential equations

$$P_i(f) = 0, \quad i = 1, \ldots, m. \tag{1}$$

*The A_n**-module associated to the system (1)** is $A_n / \sum_1^m A_n P_i$. A polynomial solution of (1) is a polynomial $f \in K[X]$ which satisfies $P_i(f) = 0$, for $i = 1, \ldots, m$. The set of all polynomial solutions of (1) forms a K-vector space.*

Theorem 1.2. *Let M be the A_n-module associated with the system (1). The K-vector space of polynomial solutions of the system (1) is isomorphic to* $\mathrm{Hom}_{A_n}(M, K[X])$.

Proof. Step1. Let f be a polynomial solution of (1). Define a map $\phi_f : A_n \to K[X]$ by the rule

$$\phi_f(D) = D(f)$$

It is easy to check that ϕ_f is an A_n-module homomorphism. Moreover, if $D \in \sum_1^m A_n P_i$, then

$$\phi_f(D) = D(f) = \sum_1^m D_i P_i(f) = 0.$$

that is $\sum_1^m A_n P_i \subseteq \ker \phi_f$. Thus ϕ_f induces an A_n-module homomorphism $\bar{\phi}_f : M \to K[X]$.

$$\begin{tikzcd} A_n \arrow[r, "\phi_f"] \arrow[d, "\pi"] & K[X] \\ M \arrow[ur, "\bar{\phi}_f"] & \end{tikzcd}$$

Step 2. Conversely, let $\psi : M \to K[X]$ be an A_n-module homomorphism. Define $f = \psi(\overline{1})$, where $\overline{1}$ is the image of 1 in M. Then for $i = 1, \ldots, m$,

$$P_i(f) = P_i(\psi(\overline{1})) = \psi(P_i(\overline{1})) = 0.$$

Thus f is a polynomial solution of (1).

It is easy to check that the maps $f \mapsto \bar{\phi}_f$ and $\psi \mapsto \psi(\overline{1})$ are inverse to each other. Hence we have established the required isomorphism. □

Definition 1.3. *Let $\mathcal{S}$ be a left A_n-module; and let $M = (P_i)$ be a finitely generated left A_n-module. We will call* $\mathrm{Hom}_{A_n}(M, \mathcal{S})$ *the solution space of M in $\mathcal{S}$ which isomorphic to K-vector space of solutions of*

$$P_i\,(f) = 0 \textit{ in } \mathcal{S}, \quad i = 1, \ldots, m.$$

§2 Microfunctions

Let $D(\epsilon)$ be the open disk of $\mathbb{C}$ of centre 0 and radius ϵ, $D'(\epsilon) = D(\epsilon)\backslash 0$ and $\mathcal{H}\,(\Omega)$ be the set of holomorphic functions in the open set $\Omega \subseteq \mathbb{C}$ viewed as an $A_1(\mathbb{C})$-module.

Proposition 2.1. *Let directed set $I = \{D(\epsilon) : \epsilon \in \mathbb{R}\}$ such that $D(\epsilon) \leq D(\epsilon')$ iif $D(\epsilon) \supseteq D(\epsilon')$, and a directed family $\{\mathcal{H}(D\,(\epsilon)) : D(\epsilon) \in I\}$. The homomorphisms $\tau_{\epsilon\epsilon'} : \mathcal{H}(D(\epsilon)) \longrightarrow \mathcal{H}\,(D\,(\epsilon'))$ are defined by restriction of holomorphic functions.*

*The elements of $\mathcal{H}_0 = \varinjlim \mathcal{H}(D(\epsilon))$ are called **germs of holomorphic functions** at 0.*

Proposition 2.2. *The **universal cover** of $D'(\epsilon)$ is the set $\tilde{D}(\epsilon) = \{z \in \mathbb{C} : \mathrm{Re}(z) < \log(\epsilon)\}$. The projection π of $\tilde{D}(\epsilon)$ on $D'(\epsilon)$ is defined by $\pi(z) = e^z$. We have the commutative diagram*

$$\begin{tikzcd} \tilde{D}(\epsilon) \arrow[d, "\pi"] & \\ D'(\epsilon) \arrow[r, hook] & D(\epsilon) \end{tikzcd}$$

Proposition 2.3. *Let $h \in \mathcal{H}(\tilde{D}(\epsilon))$. The action of a polynomial $f \in \mathbb{C}[x]$ on h is given by $f \bullet h = f\,(e^z)\,h(z)$. The operator $\partial = d/dx$ acts on h by the formula $\partial \bullet h = h'(z)e^{-z}$. Then the map*

$$\pi^* : \mathcal{H}\,(D'(\epsilon)) \longrightarrow \mathcal{H}(\tilde{D}(\epsilon))$$

defined by $\pi^*(h) = h \circ \pi$ *is an injective homomorphism of* $A_1(\mathbb{C})$ *modules.*

Definition 2.4. *Let* $\mathcal{M}_\epsilon$ *denote the quotient module* $\mathcal{H}(\tilde{D}(\epsilon))/\pi^*(\mathcal{H}(D(\epsilon)))$. *If* $D(\epsilon) \leq D(\epsilon')$, *then* $\tilde{D}(\epsilon') \subseteq \tilde{D}(\epsilon)$ *and* $\mathcal{H}(\tilde{D}(\epsilon)) \subseteq \mathcal{H}\left(\tilde{D}(\epsilon')\right)$. *This induces a homomorphism of* $A_1(\mathbb{C})$*-modules*

$$\tau_{\epsilon\epsilon'} : \mathcal{M}_\epsilon \longrightarrow \mathcal{M}_{\epsilon'}.$$

Hence $\{\mathcal{M}_\epsilon : \epsilon \in \mathbb{R}\}$ *is a directed family of* $A_1(\mathbb{C})$*-modules, and we may take its direct limit called the* ***module of microfunctions****, denoted by* $\mathcal{M}$.

Chapter VI

§1 Filtration and Associated Graded Rings and Modules

§1.1 Increasing filtration

Definition 1.1. *Let A be a ring. A **increasing filtration** of A is a sequence of subgroups $\{F_iA\}$ of A such that*

(i) $F_0A \subset F_1A \subset F_2A \subset \cdots \subset F_nA \subset \cdots \subset A$

(ii) $\bigcup_{i\geq 0} F_iA = A$.

(iii) $F_iA \cdot F_jA \subseteq F_{i+j}A$ *for all* i, j.

*A is called a **filtered ring** if it has an increasing filtration.*

Proposition 1.2. *Let A be a ring.*

1. *If $A = \bigoplus_{i\geq 0} A_i$ is a graded ring, then the sequence of subgroups $\left\{F_kA := \bigoplus_{i=0}^{k} A_i\right\}$ is an increasing filtration of A.*

2. *If $\mathcal{F} = \{F_iA\}$ is a filtration of A, then*

$$\mathrm{gr}_{\mathcal{F}} A := \bigoplus_{i\geq 0} F_{i+1}A/F_iA$$

*is a graded ring (multiplication follows from A), called the **associated graded ring of** A **associated with the filtration** $\mathcal{F}$.*

Definition 1.3. *Let A be a filtered ring with increasing filtration $\mathcal{F} = \{F_iA\}$. A left A-module M is called a **filtered left** A**-module** if it has a sequence of subgroups $\Gamma = \{\Gamma_iM\}$ such that*

(i) $\Gamma_0M \subset \Gamma_1M \subset \Gamma_2M \subset \cdots \subset \Gamma_nM \subset \cdots \subset M$

(ii) $\bigcup_{i\geq 0} \Gamma_iM = M$

(iii) $F_iA \cdot \Gamma_jM \subseteq \Gamma_{i+j}M$ *for all* i, j.

Proposition 1.4. *Let M be a left A-module where A is a filtered ring with increasing filtration $\mathcal{F} = \{F_i A\}$.*

1. *If $M = \bigoplus_{i \geq 0} M_i$ is a graded A-module, then the sequence of subgroups $\Gamma_k M := \bigoplus_{i=0}^{k} M_i$ is an increasing filtration of M.*

2. *If $\Gamma = \{\Gamma_i M\}$ is a filtration of M, then The **graded module of** M **associated with the filtration** Γ is defined by*
$$\mathrm{gr}_\Gamma M = \bigoplus_{i \geq 0} \Gamma_{i+1} M / \Gamma_i M$$
which is a graded $\mathrm{gr}_{\mathcal{F}} A$-module.

Remark. *In (1), the $\mathrm{gr}_{\mathcal{F}} M \cong M$ (as graded modules).*

But in (2), let $A = \mathbb{Z}$ with trivial filtration and $M = \mathbb{Z}_{p^2}$ with filtration $F_0 M = 0, F_1 M = p\mathbb{Z}_{p^2}, F_2 M = M, \ldots$, then $\mathrm{gr}_{\mathcal{F}} M \cong \mathbb{Z}_p^2 \ncong M$ in ${}_{\mathbb{Z}}\mathbf{Mod}$.

The functor $\mathrm{gr}(-) : {}_R\mathbf{FiltMod} \to {}_R\mathbf{GrMod}$ is not faithful

§2 Good filtration

Definition 2.1. *Let A be a filtered ring with an increasing filtration $\{F_n A\}$, and M a filtered A-module with an increasing filtration $\{F_n M\}$. The filtration of M is called a **good filtration** if $\mathrm{gr}(M)$ is finitely generated over $\mathrm{gr}(A)$.*

§3 Induced filtrations

Definition 3.1. *Let A be a filtered ring with filtration $\mathcal{F} = \{F_i A\}$ and M be a left filtered A-module with filtration $\Gamma = \{\Gamma_i\}$.*

1. *If N is a submodule of M, then the **induced filtration** on N is a filtration $\Gamma' = \{\Gamma_i N\}$ defined by*
$$\Gamma'_i := N \cap \Gamma_i.$$

   Remark. *associated graded module $\mathrm{gr}_{\Gamma'} N$ of N and graded module homomorphism $i_k : N \cap \Gamma_k / N \cap \Gamma_{k-1} \to \Gamma_k / \Gamma_{k-1}$*

2. *the **quotient filtration** $\Gamma'' = \{\Gamma''_i\}$ on M/N is defined by*
$$\Gamma''_i := \Gamma_i / (N \cap \Gamma_i).$$

   Remark. *associated graded module $\mathrm{gr}_{\Gamma''}(M/N)$ of M/N and graded module homomorphism $\pi_k : \Gamma_k M / \Gamma_{k-1} M \to (\Gamma_k M / (N \cap \Gamma_k M)) \,/\, (\Gamma_{k-1} M / (N \cap \Gamma_{k-1} M))$ (note that the right hand side is isomorphic to $\Gamma_k M / (\Gamma_{k-1} M + \Gamma_k M \cap N)$, $(A_1/B_1) \,/\, (A_2/B_2) \cong A_1 / (B_1 + A_2)$)*

Proposition 3.2. *Then there exists a short exact sequence of graded* $\mathrm{gr}_{\mathcal{F}} R$*-modules*

$$0 \longrightarrow \mathrm{gr}_{\Gamma'} N \xrightarrow{i} \mathrm{gr}_{\Gamma} M \xrightarrow{\pi} \mathrm{gr}_{\Gamma''}(M/N) \longrightarrow 0.$$

is exact.

Proof. It is clear that the each

$$0 \longrightarrow \Gamma'_k \cap \Gamma'_{k-1} \xrightarrow{i_k} \Gamma_k/\Gamma_{k-1} \xrightarrow{\pi_k} \Gamma''_k/\Gamma''_{k-1} \longrightarrow 0$$

is exact in **Ab**. Thus the proposition follows. □

§4 Dimension

Definition 4.1. *Let* M *be a finitely generated left* A_n*-module. Suppose that* Γ *is a good filtration of* M *with respect to the Bernstein filtration* $\mathcal{B}$*.*

1. *Denote by* $\chi(t, \Gamma, M)$ *the* ***Hilbert polynomial*** *of the graded module* $gr^{\Gamma} M$ *over* S_n*. We have*

$$\chi(t, \Gamma, M) = \sum_{0}^{t} \dim_k (\Gamma_i/\Gamma_{i-1}) = \dim_k (\Gamma_t).$$

   The dimension $d(M)$ *of* M *is defined the* $\deg \chi(t, \Gamma, M)$*.*

2. *Let* $a_{d(M)}$ *be the leading coefficient of* $\chi(t, \Gamma, M)$*. The* ***multiplicity*** *of* M *is* $m(M) = d! a_{d(M)}$*.*

Remark. *The dimension and multiplicity of* M *do not depend on the choice of the good filtration* Γ*.*

Example 4.2. *First let* M *be the left* A_n*-module* A_n*. The Bernstein filtration* $\mathcal{B}$ *is a good filtration of* M *and it is possible to calculate* $\chi(t, \mathcal{B}, M)$ *explicitly in this case.*

$$\chi(t, \mathcal{B}, M) = \#\left\{x^{\alpha}\partial^{\beta} : \alpha, \beta \in \mathbb{N}^n, |\alpha| + |\beta| \leq t\right\} = \binom{t+2n}{2n}$$

that is

$$\frac{(t+2n)(t+2n-1)\cdots(t+1)}{(2n)!}$$

Thus $d(A_n) = 2n$ *and* $m(A_n) = 1$*.*

Example 4.3. *Another* A_n*-module that we know very well is* $K[X] = K[x_1, \ldots, x_n]$ *with good filtration* $\Gamma_i := \{f(x_1, \ldots, x_n) : \deg f \leq i\}$*. It is easy to show that*

$$\chi(t, \mathcal{B}, k[X]) = \dim_k \Gamma_t = \binom{n+t}{n}$$

Hence $d(K[X]) = n$ *and* $m(K[X]) = 1$*.*

§4.1 Upper boundness

Theorem 4.4. *Let M be a finitely generated left A_n-module and N a submodule of M.*

1. $d(M) = \max\{d(N), d(M/N)\}$.
2. *If* $d(N) = d(M/N)$ *then* $m(M) = m(N) + m(M/N)$.

Proof. Let M be a finitely generated left A_n-module and Γ a good filtration of M with respect to $\mathcal{B}$. Let N be a submodule of M. Denote by Γ' and Γ'' the filtrations induced by Γ in N and M/N, we have an exact sequence of S_n-modules, namely

$$0 \to gr_{\Gamma'} N \to gr_{\Gamma} M \to gr_{\Gamma''} M/N \to 0.$$

Since Γ is good, $gr_{\Gamma} M$ is finitely generated. But S_n is a noetherian ring. Hence $gr_{\Gamma'} N$ and $gr_{\Gamma''}(M/N)$ are also finitely generated. Therefore both Γ' and Γ'' are good filtrations. On the other hand, since the sequence of vector spaces

$$0 \to \Gamma'_m/\Gamma'_{m-1} \to \Gamma_m/\Gamma_{m-1} \to \Gamma''_m/\Gamma''_{m-1} \to 0$$

is exact, we have that

$$\dim_k \Gamma'_m/\Gamma'_{m-1} + \dim_k \Gamma''_m/\Gamma''_{m-1} = \dim_k \Gamma_m/\Gamma_{m-1}.$$

Summing these terms for $m = 0, 1, \ldots, s$ and assuming that $s \gg 0$ one obtains

$$\chi(s, \Gamma', N) + \chi(s, \Gamma'', M/N) = \chi(s, \Gamma, M)$$

The result now follows from the properties of polynomials. □

Corollary 4.5. *Let $M_1, \ldots, M_m$ be finitely generated left A_n-modules, and $M = M_1 \oplus \cdots \oplus M_m$.*

1. $d(M) = \max\{d(M_1), \ldots, d(M_m)\}$.
2. *If* $d(M) = d(M_i)$ *for* $1 \le i \le k$, *then* $m(M) = \sum_1^k m(M_i)$.

Corollary 4.6. *Let M be a finitely generated A_n-module. Then $d(M) \le 2n$.*

Proof. Suppose that M is generated by r elements. Then there exists a surjective homomorphism $\phi : {A_n}^{\oplus r} \to M$. It follows from the theorem that $d\left({A_n}^{\oplus r}\right) = \max\{d(M), d(\ker\phi)\}$, thus $d(M) \le d\left({A_n}^{\oplus r}\right) = 2n$. □

Corollary 4.7. *Let I be a non-zero left ideal of A_n. Then $d(A_n/I) \le 2n - 1$.*

Proof. First consider the case of a cyclic left ideal. Let $d \in A_n$, and put $I = A_n d$. Then we have an exact sequence

$$0 \to A_n \xrightarrow{\theta} A_n \to A_n/A_n d \to 0$$

where the map θ is defined by $\theta(a) = ad$, for every $a \in A_n$. Suppose, by contradiction, that $d\left(A_n/A_n d\right) = 2n$. Then

$$m\left(A_n\right) = m\left(A_n\right) + m\left(A_n/A_n d\right).$$

Since $m\left(A_n\right) = 1$ and the multiplicity is a positive number, this equation is impossible. Hence $d\left(A_n/A_n d\right) \leq 2n - 1$.

Now for the general case. Let I be a non-zero left ideal of A_n and choose $0 \neq d \in I$. Since $A_n d \subseteq I$, we have that A_n/I is a quotient of $A_n/A_n d$. Since the latter has dimension $\leq 2n - 1$, so does A_n/I. $\square$

§4.2 Lower boundness

Theorem 4.8 (Bernstein's Inequality)**.** *If M is a finitely generated non-zero left A_n-module, then $d(M) \geq n$.*

Chapter VII

Holonomic Modules

§1 Definition

Definition 1.1. *A finitely generated left A_n-module is* ***holonomic*** *if it is zero, or if it has dimension n.*

Proposition 1.2. *Let n be a positive integer.*

1. *Submodules and quotients of holonomic A_n-modules are holonomic.*
2. *Finite direct sums of holonomic A_n-modules are holonomic.*

Corollary 1.3. *Finitely generated torsion A_1-modules are holonomic.*

Proposition 1.4. *Holonomic A_n-modules are torsion modules.*

§2 Basic properties

Theorem 2.1. *Holonomic modules are artinian.*

Proof. Let M be a holonomic left A_n-module. Suppose that M has a descending chain of submodules

$$M = N_0 \supseteq N_1 \supseteq N_2 \supseteq \cdots \supseteq N_r$$

it follows that $m\left(N_i\right) = m\left(N_{i+1}\right) + m\left(N_i/N_{i+1}\right)$. Putting these together, we get that

$$m(M) = \sum_{0}^{r-1} m\left(N_i/N_{i+1}\right) + m\left(N_r\right) \geq r$$

Hence M cannot have a descending chain of more than r submodules. In particular M cannot have an infinite descending chain. $\square$

Remark. *Noted that the regular left module A_n is not artinian (ring A_n is not left Artinian). It is easy to construct an infinite descending chain; take for instance*

$$A_n x_n \supseteq A_n x_n^2 \supseteq A_n x_n^3 \supseteq \dots .$$

Corollary 2.2. *Every holonomic A_n-module has finite length that cannot exceed its multiplicity.*

Corollary 2.3. *Every irreducible holonomic A_n-module has multiplicity* 1.

Chapter VIII

Lemma 0.1. *Let Ω and Γ be both good filtration of finitely generated left A_n-module M. Then*

$$\mathrm{Rad}\left(\mathrm{Ann}\left(\mathrm{gr}_\Gamma M\right)\right) = \mathrm{Rad}\left(\mathrm{Ann}\left(\mathrm{gr}_\Omega M\right)\right)$$

in S_n.

Definition 0.2. *The ideal*

$$\mathcal{I}(M) = \mathrm{rad}(\mathrm{ann}(M, F))$$

*is called the **characteristic ideal** of M. Thus the affine variety*

$$\mathrm{Ch}(M) = \mathcal{Z}(\mathcal{I}(M)) \subseteq \mathbb{C}^{2n}$$

*is called the **characteristic variety** of M.*

Remark. *Since $\mathcal{I}(M)$ is a homogeneous ideal, the variety $\mathrm{Ch}(M)$ is homogeneous. This means that if $p \in \mathrm{Ch}(M)$ and $\lambda \in \mathbb{C}$, then $\lambda p \in \mathrm{Ch}(M)$. Note that $\mathrm{Ch}(M)$ is a subvariety of $\mathbb{C}^{2n}$, since S_n is a polynomial ring in $2n$ variables!*

Proposition 0.3. *Let M be a finitely generated left A_n-module and N a submodule of M. Then*

$$\mathrm{Ch}(M) = \mathrm{Ch}(N) \cup \mathrm{Ch}(M/N).$$

Definition 0.4. *Let J be an ideal of $S_n = \mathbb{C}\left[y_1, \ldots, y_{2n}\right]$ and put algebric set $V = \mathcal{Z}(J)$.*

1. *If p is a point of V then the **Zariski tangent space** of V at p is the linear subspace of $\mathbb{C}^{2n}$ defined by the equations*

$$\sum_1^{2n} \frac{\partial F}{\partial y_i}(p) y_i = 0$$

   for all $F \in J$. This space is denoted by $T_p(V)$; it is a complex vector subspace of $\mathbb{C}^{2n}$.

2. *The **dimension** of V equals $\inf\{\dim_{\mathbb{C}} T_p(V) : p \in V\}$.*

Remark. *Actually one need not look at every point of* V*. If* p *is a non-singular point of* V*, then* $\dim(V) = \dim_{\mathbb{C}} T_p(V)$*. This is equivalent to the definition in terms of heights of prime ideals, see [Hartshorne 77, Ch. 1, Exercise 5.10].*

Theorem 0.5. *Let* M *be a finitely generated left module over* A_n*. Then* $\dim \mathrm{Ch}(M) = d(M)$.

Chapter IX

Recall that we have the notations $X = k^n, Y = k^m, k[X] = k[x_1, \ldots, x_n], Y = k[y_1, \ldots, y_m]$. Let $F : X \to Y$ be a polynomial map, its comorphism $F^\sharp : k[Y] \to k[X]$ is an algebra homomorphism defined by

$$F^\sharp(\varphi) := \varphi \circ F = \varphi\left(F_1, \ldots, F_m\right)$$

§1 Tensor Product

Proposition 1.1. *Let A be a k-algebra and let A_1 and A_2 be subalgebras of A. Suppose that*

(i) $A = A_1 A_2$,

(ii) $[A_1, A_2] = 0$,

(iii) there exist k-bases $\{a_i\}$ and $\{b_j\}$ of A_1 and A_2, respectively, such that $\{a_i b_j\}$ is a k-basis of A.

Then $A \cong A_1 \otimes_k A_2 := A_1 \widehat{\otimes} A_2$ as k-algebras.

Corollary 1.2. *The following isomorphisms are induced by the multiplication map:*

1. $k[X] \otimes_k k[Y] \cong k[X, Y]$.
2. $A_m \otimes_k A_n \cong A_{m+n}$.

Definition 1.3. *Let A, B be k-algebras. Suppose that M is a left A-module and that N is a left B-module. Then $M \otimes_k N$ is a left $A \otimes_k B$-module with action defined by*

$$(a \otimes b)(u \otimes v) = au \otimes bv.$$

We use the notation $A \widehat{\otimes} B$ and $M \widehat{\otimes} N$ to denote the tensor products over k when we want to stress the fact that we are considering the module structure defined above.

Proposition 1.4. *If M and N are finitely generated modules, then so is $M \widehat{\otimes} N$.*

Theorem 1.5. *Let M be a finitely generated left A_m-module and N a finitely generated left A_n-module. Then*

1. $d(M\widehat{\otimes}N) = d(M) + d(N)$,

2. $m(M\widehat{\otimes}N) \leq m(M)m(N)$.

§2 Inverse Image

Definition 2.1. *Let polynomial map*

$$F : X \to Y$$

be given. Its comorphism is the algebra homomorphism

$$F^\sharp : k[Y] \to k[X]$$

defined by $\varphi \mapsto \varphi \circ F$. It may be used to make $k[Y]$-modules M into $k[X]$-modules:

$$F^\star : {}_{k[Y]}\mathbf{Mod} \to {}_{k[X]}\mathbf{Mod}$$

*defined by $F^\star(M) := k[X] \otimes_{k[Y]} M$ called the **inverse image** of M by the polynomial map F.*

Definition 2.2. *If M is a left A_m-module, it follows that M is also a left $k[Y]$-module. We now turn $k[X]$-module $F^\star(M)$ into a A_n-module*

$$F^\star = {}_{A_m}\mathbf{Mod} \to {}_{A_n}\mathbf{Mod}$$

by giving the recipe for the action

$$x_i\,(q \times u) = (x_i q) \otimes u$$

$$\partial_{x_i}(q \otimes u) = \partial_{x_i}(q) \otimes u + \sum_{k=1}^{m} q\partial_{x_i}\,(F_k) \otimes \partial_{y_k} u$$

for $i = 1, \ldots, n$.

Remark. *It is well defined and satisfies the relations*

$$[\partial_{x_i}, x_j]w = \delta_{ij}w, \quad [\partial_{x_i}, \partial_{x_j}]w = 0, \quad [x_i, x_j]w = 0$$

for all $w \in F^\star(M)$.

Theorem 2.3. *Let $F : X \to Y$ and $G : Y \to Z$ be polynomial maps and M be an A_r-module. Then*

$$F^\star G^\star M \cong (GF)^\star M$$

Corollary 2.4. *Let $F : X \to X$ be a polynomial isomorphism and G its inverse. If M is a left A_n-module, then $F^\star G^\star M \cong M$.*

Theorem 2.5. *Let $F : X \to Y$ be a polynomial map. If M is a holonomic A_m-module, then $F^\star M$ is a holonomic A_n-module.*

§2.1 Projections

Example 2.6. *The second example is the projection $\pi : X \times Y \to Y$ onto the second coordinate. The comorphism $\pi^\sharp$ maps a polynomial $p \in k[Y]$ to itself, but considered as an element of $k[X, Y]$.*

Let M be a module over $k[Y]$, then

$$\pi^\star(M) = k[X, Y] \otimes_{k[Y]} M \cong k[X] \widehat{\otimes} M$$

as $k[X, Y]$-modules by

$$q_\alpha x^\alpha \otimes u \longmapsto q_\alpha \otimes q_\alpha u$$

Theorem 2.7. *Let M be a finitely generated left A_m-module and π the projection defined above. Then:*

1. *$\pi^\star M$ is a finitely generated A_{m+n}-module.*
2. *$d(\pi^\star M) = n + d(M)$.*
3. *$m(\pi^\star M) \leq m(M)$.*

§2.2 Embeddings

Definition 2.8. *We begin by considering the standard embedding*

$$\iota : X \to X \times Y$$

defined by $\iota(X) = (X, 0)$, where 0 denotes the origin of Y. The comorphism

$$\iota^\sharp : K[X, Y] \to K[X]$$

is defined by $\iota^\sharp(g(X, Y)) = g(X, 0)$ (gives a isomorpjism $k[X, Y]/(Y) \cong k[X]$). It may be used to make $k[X, Y]$-modules M into a $k[X]$-module:

$$\iota^\star : {}_{k[X,Y]}\mathbf{Mod} \to {}_{k[X]}\mathbf{Mod}$$

defined by $\iota^\star(M) := k[X] \otimes_{k[X,Y]} M$.

Now consider left A_{n+m}-module M, the A_n-modules $\iota^\star(M)$ is defined by the action

$$\partial_{x_i}(q \otimes u) = \frac{\partial q}{\partial x_i} \otimes u + q \otimes \partial_{x_i} u$$

Proposition 2.9. *Let M be a left A_{n+m}-module. Then*

$$k[X] \otimes_{k[X,Y]} M \cong {}_{A_n}M/\,(Y)\,{}_{A_n}M$$

as A_n-modules.

Proof. If $u \in M$, let $\bar{u}$ denote its image in $M/(Y)M$. Define the map

$$\phi : \iota^\star M \to M/(Y)M$$

by $\phi(q \otimes u) = q\bar{u}$. It is a homomorphism of $K[X]$-modules. It suffices to show that the action of the derivations $\partial_{x_1}, \ldots, \partial_{x_n}$ is compatible with ϕ and ϕ is injective. □

§3 Direct Image

Definition 3.1. *Let $F : X \to Y$ be a polynomial map. Let M be a left A_m-module.*

1. *The A_n-A_m bimodule*

$$D_{X\to Y} := k[X] \otimes_{k[Y]} A_m$$

   Noted that $F^\star M \cong D_{X\to Y} \otimes_{A_m} M$

2. *Let N be a right A_n-module. The **direct image** of N by F is the right A_m-module defined by*

$$N \otimes_{A_n} D_{X\to Y}$$

Theorem 3.2. *Let $\iota : X \to X \times Y$ be the standard embedding. If M is a finitely generated left A_n-module, then:*

1. *$\iota_\star M$ is a finitely generated left A_{m+n}-module.*
2. *$d\,(\iota_\star M) = m + d(M)$.*
3. *$m\,(\iota_\star M) \leq m(M)$.*

2.3 Theorem. Let $F : X \to Y$ be a polynomial map. If M is a holonomic A_n-module, then $F_\star M$ is a holonomic A_m-module.